ACTIVITY

(Multiple Interactive Learning Algorithm Activity)

Game Based Activity

SLOT A

MLA0403– DEEP LEARNING

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CNN ARCHITECTURE RACE

ResNet vs AlexNet

A head-to-head look at two landmark convolutional neural network designs — and which one fits your next computer-vision project.

Convolutional Neural Network

Built for computer vision

Learns visual features automatically

What Makes a CNN Tick

The building blocks behind AlexNet and ResNet alike

Convolutional Neural Network

"CNN" — a deep learning architecture purpose-built for visual data.

Image classification & vision

Mainly used for image classification and broader computer-vision tasks.

Automatic feature learning

CNNs learn visual features directly from raw pixels — no manual engineering.

The objective

Identify which architecture — AlexNet or ResNet — best fits a given application.

TIMELINE

2012

AlexNet

ImageNet breakthrough — sparks the deep-learning era

2015

ResNet

Residual connections enable ultra-deep networks

Both classify images into categories — the difference is how deep, accurate, and efficient they are.

AlexNet

2012 · The breakthrough

One of the major breakthroughs in deep learning — it achieved excellent results in the ImageNet competition and helped ignite the modern deep-learning boom.

8 trainable layers

~60M parameters

15.3% top-5 error (ILSVRC '12)

Key Characteristics

Conv + pooling + fully connected

Stacks convolutional layers, pooling layers, and fully connected layers in sequence.

ReLU activation

Uses ReLU activation for faster, more stable training than earlier sigmoid/tanh nets.

Dropout regularization

Uses dropout to reduce overfitting on the fully connected layers.

Great for learning fundamentals

Its simple, linear structure makes it ideal for understanding how CNNs work.

Solid for basic classification

Well suited to basic image-classification tasks that don't demand extreme accuracy.

Key Characteristics

Residual (skip) connections

Shortcuts let gradients flow through very deep stacks without vanishing.

Extreme depth, stable training

Scales to 34, 50, 101, even 152 layers while remaining trainable.

State-of-the-art accuracy

Delivered a step-change accuracy jump over AlexNet-era architectures.

Batch normalization

Normalizes activations between layers, speeding convergence at depth.

Built for demanding vision tasks

The go-to choice when an application needs maximum classification accuracy.

ResNet

2015 · The deep leap

Introduced residual ("skip") connections that let networks go far deeper than before — without the accuracy loss that used to come with extra depth.

layers in the deepest variant	152
top-5 error (ILSVRC '15)	3.57%
to beat human-level accuracy	1st

Choosing Your Fighter

AlexNet vs ResNet, side by side

Category	AlexNet	ResNet
Introduced	2012	2015
Depth	8 layers	Up to 152 layers
Key innovation	ReLU + dropout	Residual (skip) connections
Top-5 error	15.3%	3.57%
Best for	Learning fundamentals, light tasks	High-accuracy production vision

Verdict: Pick AlexNet to learn CNN fundamentals or run lightweight classification. Pick ResNet when the application demands top-tier accuracy on complex, large-scale image data.